

Tuning collective anion motion enables superionic conductivity in solid-state halide electrolytes

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Halides of the family Li_3MX_6 ($\text{M} = \text{Y}, \text{In}, \text{Sc}$ and so on, $\text{X} = \text{halogen}$) are emerging solid electrolyte materials for all-solid-state Li-ion batteries. They show greater chemical stability and wider electrochemical stability windows than existing sulfide solid electrolytes, but have lower room-temperature ionic conductivities. Here we report the discovery that the superionic transition in Li_3YCl_6 is triggered by the collective motion of anions, as evidenced by synchrotron X-ray and neutron scattering characterizations and ab initio molecular dynamics simulations. Based on this finding, we used a rational design strategy to lower the transition temperature and thus improve the room-temperature ionic conductivity of this family of compounds. We accordingly synthesized $\text{Li}_3\text{YCl}_x\text{Br}_{6-x}$ and $\text{Li}_3\text{GdCl}_3\text{Br}_3$ and achieved very high room-temperature conductivities of 6.1 and 11 mS cm^{-1} for $\text{Li}_3\text{YCl}_{4.5}\text{Br}_{1.5}$ and $\text{Li}_3\text{GdCl}_3\text{Br}_3$, respectively. These findings open new routes to the design of room-temperature superionic conductors for high-performance solid batteries.

All-solid-state lithium-ion batteries (ALSOLIBs) are a promising next-generation energy storage technology owing to their high safety and energy density as well as their versatility of applications^{1,2}. As a key component of ALSOLIBs, solid electrolytes have been extensively explored in recent years and a variety of materials have been reported, including garnet-type ($\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$)^{3,4}, NASICON (Na super ionic conductor)-type ($\text{Li}_{1.3}\text{Al}_{0.3}\text{Ti}_{1.7}(\text{PO}_4)_3$)^{5,6} and argyrodite-type ($\text{Li}_{6-x}\text{PS}_{5-x}\text{X}_x$, $\text{X} = \text{Cl}$ and Br)⁷ materials, as well as $\text{Li}_7\text{P}_3\text{S}_{11}$ (refs. 8,9) and $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$ (refs. 2,10). However, the ideal solid electrolyte with all of the properties desired for the large-scale commercialization of ALSOLIBs, including high room-temperature (RT) ionic conductivity, wide electrochemical window, good chemical stability, good mechanical properties and low cost, is yet to be discovered¹¹. The Li_3MX_6 ($\text{M} = \text{Y}, \text{Sc}, \text{In}$ and so on, $\text{X} = \text{halogen}$)^{12–14} and $\text{Li}_2\text{M}'\text{X}_6$ ($\text{M}' = \text{Zr}$ and Hf , $\text{X} = \text{halogen}$)^{15–19} halide materials have recently attracted considerable attention due to their

outstanding advantages in cathode stability compared with sulfides and in ionic conductivity compared with oxides. However, the RT conductivities of these halides are still commonly one order of magnitude lower than those of sulfides, which is limiting their application and demanding improvement.

Asano et al. reported a high RT conductivity (σ_{RT}) of 0.51 mS cm^{-1} for Li_3YCl_6 (LYC) and an activation energy (E_a) of 0.40 eV, and a higher σ_{RT} of 1.7 mS cm^{-1} for Li_3YBr_6 (LYB)¹². However, the ab initio molecular dynamics (AIMD) simulations conducted by Mo and co-workers suggested that much higher σ_{RT} values (14 mS cm^{-1}) and much lower E_a values (0.19 eV) are possible for LYC with the $P3m1$ structure²⁰. In a recent study on $\text{Li}_3\text{YCl}_3\text{Br}_3$ (ref. 21), which is isostructural with LYB, higher σ_{RT} (7.2 mS cm^{-1}) and lower E_a (0.25 eV) values were achieved, which motivated us to explore whether higher ($>0.51 \text{ mS cm}^{-1}$) σ_{RT} values can be experimentally achieved for compounds with the LYC structure.

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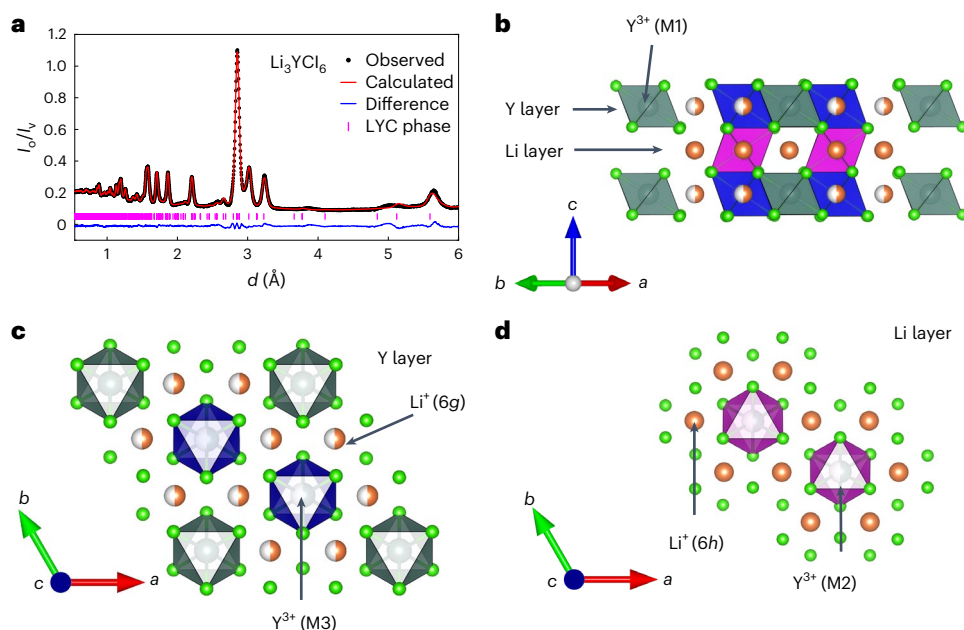


Fig. 1 | Crystal structure of LYC. **a**, Rietveld refinement against ND data (from detector bank 2, centred at $2\theta = 31^\circ$). I_0 and I_v are the neutron scattering signal of the sample and the neutron scattering signal from the 6 mm diameter vanadium rod, respectively. Bragg reflections are shown as magenta markers (LYC phase). **b**, Crystal structure of LYC viewed from the $[110]$ direction. **c,d**, Crystal structure

viewed from the $[001]$ direction showing the Y layer (**c**) and the Li layer (**d**). Y at the $1a$ site is shown in dark cyan, Li^+ at the $6h$ site (fully occupied) is shown in orange and the partially occupied $2d$ sites of Y in the Y and Li layers are shown in blue and magenta, respectively.

In this study, we systematically analysed LYC samples by synchrotron in situ X-ray diffraction (XRD), electrochemical impedance spectroscopy (EIS) and in situ neutron diffraction (ND) at different temperatures. A continuous transition to super-ionic conducting state, that is, type II superionic transition (SIT), with significantly lower E_a values above the transition temperature (T_c) was observed, which was found to be related to changes in the diffusion pathways of Li triggered by the collective motion of the anions. Guided by this finding, two series of compounds, $\text{Li}_3\text{YCl}_{6-x}\text{Br}_x$ and $\text{Li}_3\text{GdCl}_{6-x}\text{Br}_x$, were designed in an attempt to achieve lower T_c and higher RT conductivity. The compounds were successfully synthesized, electrochemically evaluated and characterized. The T_c was lowered to 70°C for $\text{Li}_3\text{YCl}_{4.5}\text{Br}_{1.5}$ and ionic conductivities of 100 mS cm^{-1} at 80°C and 6.1 mS cm^{-1} at RT were achieved. An even lower T_c of -10°C and a higher RT conductivity of 11 mS cm^{-1} were achieved for $\text{Li}_3\text{GdCl}_3\text{Br}_3$.

Results and discussion

SIT of LYC

LYC samples were synthesized by hot pressing ball-milled LYC powder at 200°C and characterized by XRD and ND. LYC has been reported to crystallize in different polymorphs^{12,22–24} with the trigonal $P\bar{3}m1$ structure most commonly observed during solid-state syntheses. Figure 1a shows the ND pattern of the obtained LYC, which can be described as a $P\bar{3}m1$ structure with a hexagonal close-packed anion sublattice with stacking sequences of ABABAB. The average structure of the $P\bar{3}m1$ phase can be described as an alternate stacking of two types of layer (Fig. 1b). Both layers contain LiCl_6 , YCl_6 and VCl_6 (where V represents a vacancy) octahedra. Y^{3+} occupies three different octahedral sites: one $1a$ site (M1) and two $2d$ sites (M2 and M3)²⁵. As shown in Fig. 1c,d, the $1a$ site and one of the $2d$ sites (M3) lie within the same layer (hereafter denoted the Y layer). It is worth noting that the distance between the centres of the face-shared YCl_6 octahedra (between the M2 and M3 sites) is only $\sim 3\text{ \AA}$, inducing large electrostatic repulsion and energetically unfavourable states if M2 and M3 sites are simultaneously occupied. Therefore, it is likely that the partial Y occupancies refined at both sites are due to the differences in the a – b plane arrangements of

the Y vacancy ordering rather than the stacking disorder along the c direction. This has further been confirmed by the neutron pair distribution function (nPDF; Supplementary Fig. 1), where no positive peak is observed at around $3\text{ \AA}$. The other $2d$ site in which Y locates is in the adjacent layer, which is a Li-rich layer (hereafter denoted the Li layer). Li^+ occupies two octahedral sites: $6h$ sites in the Li layer, fully occupied by Li^+ , and $6g$ sites in the Y layer, partially occupied by Li^+ at RT. The $6g$ and $6h$ sites are connected by face-shared LiCl_6 octahedra, forming a long-range diffusion channel for Li^+ along the c axis. Furthermore, the octahedral (O) sites are also broadly proposed to connect via two tetrahedral (T) sites to form two-dimensional (2D) diffusion pathways (O–T–O pathway) along the a – b plane²⁰.

We then measured the conductivity of LYC pellets in a broader temperature range than previously reported¹². The Arrhenius plot of the conductivity of LYC is shown in Fig. 2a. LYC has an ionic conductivity of $1.4 \times 10^{-4}\text{ S cm}^{-1}$ at 25°C with an E_a of 0.70 eV below 130°C . However, the slope is much flatter at higher temperatures with a greatly reduced E_a of 0.22 eV , showing a type II SIT and indicating significantly facilitated diffusion via a different mechanism above T_c . Such a SIT in halides has also been computationally reported by Ong and co-workers²⁶ and Wang et al.²⁷. This low E_a is close to the values obtained by Mo et al.²⁰ in their AIMD simulations. Extrapolating the Arrhenius curve from T_c to RT, the expected σ_{RT} is 29 mS cm^{-1} , which also agrees with the AIMD simulations²⁰. Apparently, the steep Arrhenius slope below T_c is the major reason for the discrepancy between previous experimentally measured ionic conductivities and theoretical predictions.

To understand the origin of the SIT, we investigated the structural change in LYC by in situ synchrotron XRD from 20 to 200°C . The XRD patterns in Supplementary Fig. 2 show broad diffuse scattering peaks at around 5.0 and $3.8\text{ \AA}$ throughout the temperature range, in good agreement with a previous report²⁸. However, no significant phase transition is observed in either the heating or cooling process. This indicates that the SIT observed in the EIS analysis is not caused by specific arrangements of the Y^{3+} vacancy ordering pattern or any long-range symmetry breaking.

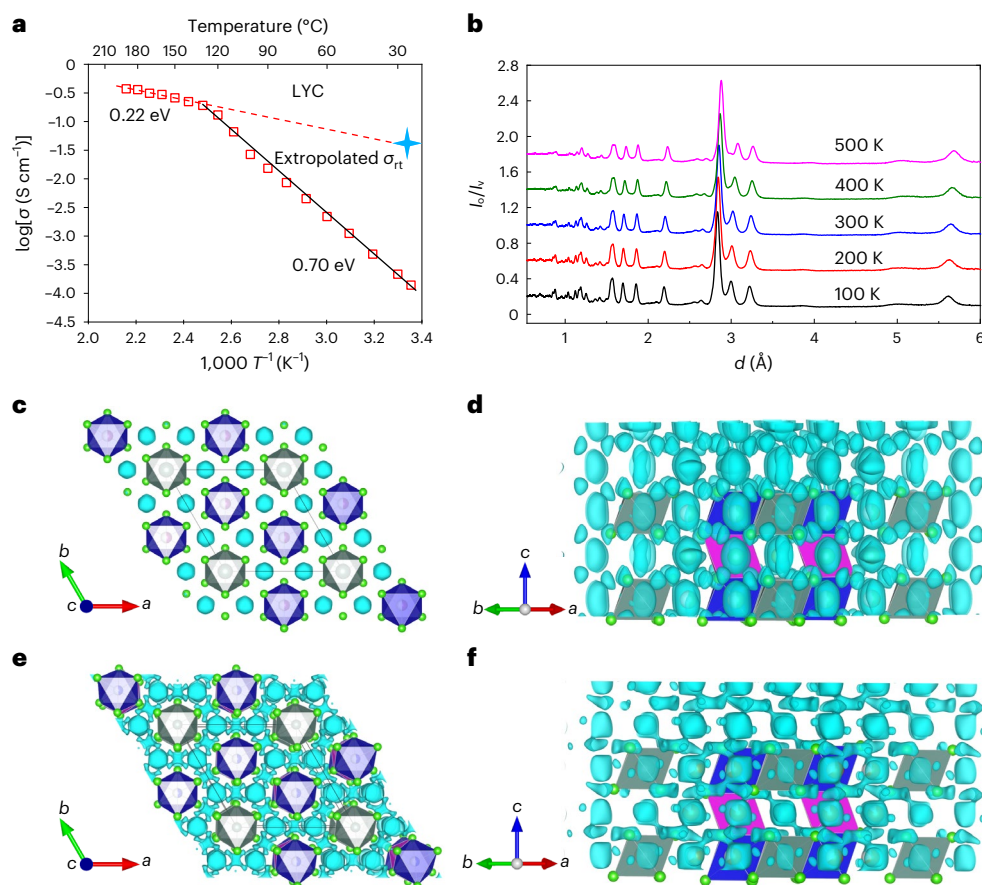


Fig. 2 | Ionic conductivity and Li diffusion paths of LYC visualized by FDMs generated from ND patterns at various temperatures. **a**, Arrhenius plot of the conductivity of LYC. **b**, ND patterns of LYC collected at different temperatures. **c,d**, FDMs generated from the ND pattern of Li_3YCl_6 at 100 K with a cut-off of

$-0.035 \text{ fm } \text{\AA}^{-3}$ (**c**) and $-0.02 \text{ fm } \text{\AA}^{-3}$ (**d**). **e,f**, FDMs generated from the ND pattern of Li_3YCl_6 at 500 K with a cut-off of $-0.03 \text{ fm } \text{\AA}^{-3}$ (**e**) and $-0.02 \text{ fm } \text{\AA}^{-3}$ (**f**). The projections along the c axis are shown in Supplementary Figs. 3 and 4.

We further investigated the structural changes by variable-temperature ND. As shown in Fig. 2b, the diffraction patterns of LYC maintain similar Bragg reflections at different temperatures despite the obvious shifts in peak positions due to thermal expansion. Fourier difference maps (FDMs; $F_{\text{calc}} - F_{\text{obs}}$, where F_{calc} and F_{obs} represent calculated and observed structure factors, respectively) were generated from the refined structure (space group $P3m1$) using the ND data collected at 100 and 500 K (Fig. 2 and Supplementary Figs. 3 and 4). At 100 K, very clear residual Li scattering length density can be seen on the two major Li sites $6g$ and $6h$ (Fig. 2c and Supplementary Fig. 3), while no noticeable density can be seen between these two sites, either within the a - b plane or along the c axis (with a cut-off of $-0.035 \text{ fm } \text{\AA}^{-3}$). Further lowering the residual density cut-off to $-0.02 \text{ fm } \text{\AA}^{-3}$ shows the connection along the c axis, but still without any obvious connection within the a - b plane (Fig. 2d). This indicates that Li^+ diffusion is likely one dimensional (along the c axis) with a relatively high energy barrier, which is consistent with the high E_a value determined by EIS (0.70 eV). In clear contrast, at 500 K (Fig. 2e), the residual Li^+ densities on the $6g$ sites are clearly connected, forming hexagonal rings around the M1Y site within the a - b plane at a threshold of $-0.03 \text{ fm } \text{\AA}^{-3}$. These rings are further connected to form 2D diffusion pathways at a threshold of $-0.02 \text{ fm } \text{\AA}^{-3}$ (Fig. 2f), indicating facilitated 2D diffusion channels within the a - b plane.

The Rietveld refinements of the ND data are shown in Fig. 3 and Supplementary Tables 1–5. Figure 3a shows the lattice parameters. The increase in the $2c/a$ ratio confirms the anisotropic thermal expansion. In previous reports, the Li sites in the Li layer ($6g$) were often assumed

to be fully occupied in the $P3m1$ structure²⁹. However, our refinements show partial occupancy of the $6g$ sites and full occupancy of the $6h$ sites below 400 K (Fig. 3b). A detailed inspection of the refined structure revealed that the Li^+ coordination environment is strongly correlated with the collective motion of Cl^- anions. From 100 to 400 K, there is only a limited change in the Li^+ occupancy and a minor change in volume of the YCl_6 octahedra (Fig. 3c). The deformation of the YCl_6 octahedra can be evaluated on the basis of the $\text{Cl}-\text{Y}-\text{Cl}$ angles: φ_{11} in M1Cl_6 , φ_{22} and φ_{33} in M3Cl_6 (Fig. 3d), and $\varphi_{22'}$ and $\varphi_{33'}$ in M2Cl_6 (Fig. 3e). The slight increase in the φ_{11} angle ($\text{Cl}-\text{M1}-\text{Cl}$) suggests that the M1Cl_6 octahedra are slightly compressed along the c axis (Fig. 3f). The neighbouring face-sharing M2Cl_6 - M3Cl_6 chains are found to rock towards the opposite direction (Supplementary Video 1). This correlated anion motion leads to only a slight increase in the intralayer $\text{Cl}-\text{Cl}$ distances (the edge between two neighbouring LiCl_6 octahedra within the a - b plane, noted as S12 and S33 in Fig. 3d). At 500 K, an abrupt elongation of the M1Cl_6 and M3Cl_6 octahedra along the c axis emerges (Fig. 3f). Accordingly, a drastic shrinking of the φ_{22} and φ_{33} ($\text{Cl}-\text{Y3}-\text{Cl}$) angles (Fig. 3f) and a drastic increase in the S12 and S33 (intralayer $\text{Cl}-\text{Cl}$) distances (Fig. 3g) were identified, which together indicate a ‘breathing’ type of correlated motion of the M1Cl_6 and M3Cl_6 octahedra and, in turn, a drastic expansion of the diffusion channel bottleneck within the a - b plane among the $6g \text{ Li}^+$ sites (in the $z = 0$ plane, that is, the Y layer). The facilitation effect of the mode shift can also be seen from the change in the area of the diffusion bottleneck windows (BN1 and BN3), as shown in Supplementary Fig. 5. In contrast, as shown in Supplementary Fig. 6a, the M2Cl_6 octahedra become compressed at

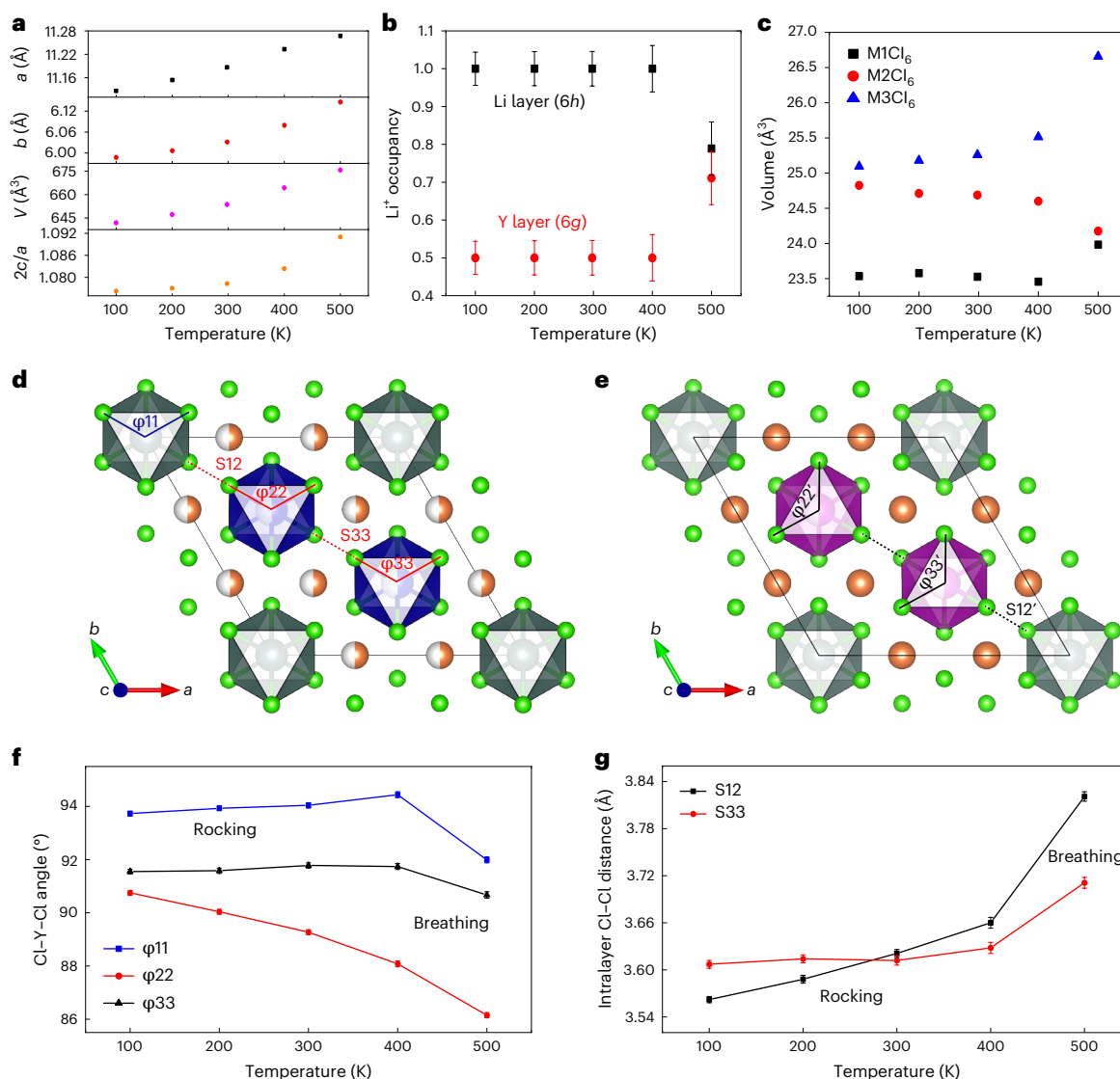


Fig. 3 | Collective motion of anion ligands and its influence on Li-ion diffusion behaviour. **a**, Lattice parameters and volume (V) extracted from Rietveld refinement of the ND patterns of LYC at various temperatures. **b**, Li occupancy at the $6g$ and $6h$ sites at various temperatures. The $6h$ sites are fully occupied below 400 K and become partially occupied at 500 K. **c**, Change in volume of $M1Cl_6$, $M2Cl_6$ and $M3Cl_6$ at various temperatures. **d, e**, Schematics of the Li-ion diffusion

bottleneck in the Y layer (**d**) and Li layer (**e**) in LYC. **f**, Evolution of the Cl–Y–Cl angle in $M1Cl_6$ and $M3Cl_6$ octahedra with temperature. **g**, Evolution of the Li-ion bottleneck S12 and S33 in the Y layer with temperature. The error bars represent the ‘estimated standard deviations’ of the corresponding refined parameters, which are a measure of the irreducible minimum in the uncertainty of the refined parameters during Rietveld refinement.

500 K (compared with the structure at 400 K) and the in-plane diffusion channel bottleneck does not change much or even shrinks slightly (S12' and S33' in Supplementary Fig. 6b) for Li^+ at the $6h$ sites (in the $z = 0.5$ plane, that is, the Li layer). This finding suggests that the 2D Li^+ diffusion channels with a low energy barrier likely open up for Li^+ only at the $6g$ sites (or $z = 0$ plane) when the temperature increases from 400 to 500 K, while in-plane Li^+ hopping is still difficult within the $z = 0.5$ plane. This is fully consistent with the results of the FDMs (Fig. 2e,f). Taken together, the observed SIT above 130 °C in LYC is very likely due to the opening up of the 2D Li^+ diffusion channels, which is rooted in the switch from the ‘rocking’ to the ‘breathing’ mode of motion of the MCl_6 units. This transition has been further confirmed by the in situ nPDF investigation (Supplementary Fig. 1). The Y–Cl bond is very stiff, as evidenced by the nearly unchanged Y–Cl bond lengths. However, the correlation between neighbouring Cl–Cl and the second-shell Y–Cl pairs becomes much weaker or even entirely uncorrelated above 400 K, suggesting that anion motion is likely to induce much wider diffusion channels for Li^+ hopping.

This transition is also confirmed by the AIMD simulations presented in Fig. 4a, where a leap in the average distance is observed for both S12 and S33 between 350 and 400 K. Ionic diffusion within the a – b plane is inactive at the low temperature of 350 K (Fig. 4b), which agrees well with the ND analysis, but becomes active at high temperatures, for example, 600 K (Fig. 4c). In addition to the leap in the average S12 and S33 distances, a broader distribution of the S12 and S33 distances and an asymmetric tail towards larger distances can also be seen, which also indicates that the bottleneck of the diffusion channel is opened up over a longer period of time at higher temperatures (Fig. 4d,e).

Tuning T_c by Br doping in LYC

The impact of anion motion on fast ionic conduction has also been discussed previously^{30,31}, which implies that this may be an important factor to tune ionic kinetics. The finding of a SIT in LYC inspired us to develop a design strategy to enhance the properties of this electrolyte material: a much-improved RT ionic conductivity can be achieved by tuning T_c to RT or lower temperatures, which may be realized by

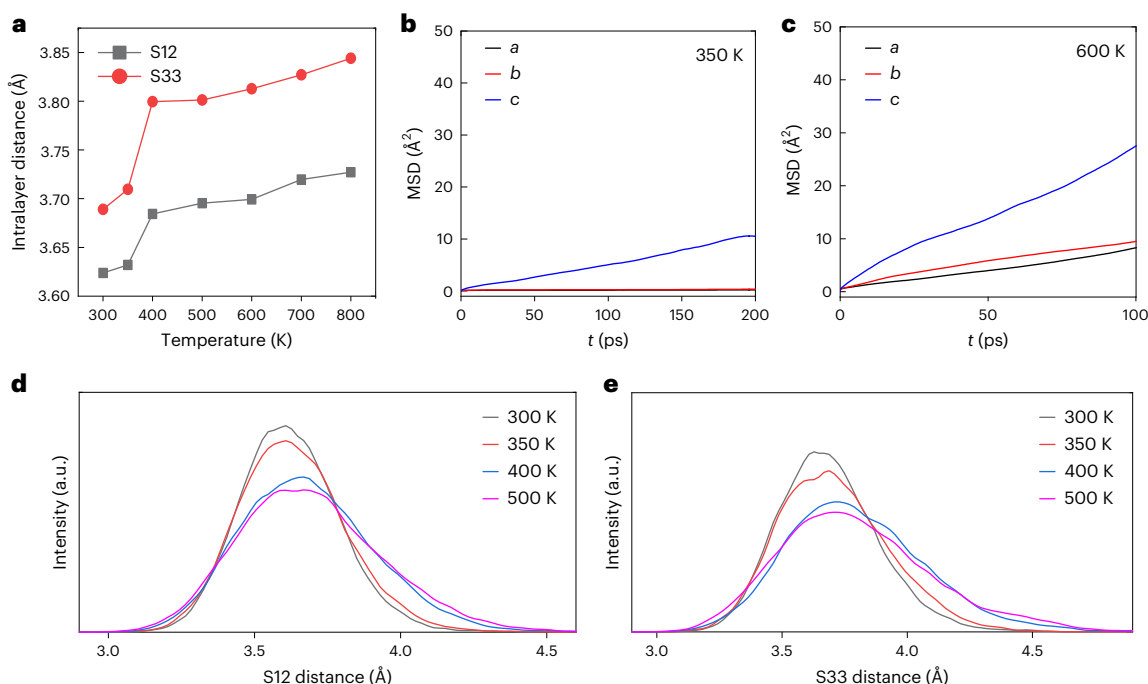


Fig. 4 | AIMD simulations of LYC at different temperatures. **a**, Evolution of the average intralayer Cl–Cl distances S12 and S33 with temperature. **b,c**, Mean square displacement (MSD) of Li⁺ in each direction at 350 K (**b**) and 600 K (**c**). **d,e**, Intralayer Cl–Cl distance distributions for S12 (**d**) and S33 (**e**) at different temperatures.

tuning the elementary composition and bonding in LYC. The ‘breathing’ mode is likely related to the properties of the metal–halogen bond. We hypothesized that with the lower electronegativity and higher polarizability of Br compared with Cl, partly replacing Cl with Br may promote the ‘breathing’ mode over the ‘rocking’ mode and thus can lower the T_c . A previous study of halide solid electrolytes with the $C2/m$ space group (LYB structure) also suggested that anion mixing may lower the migration barrier²¹. We therefore designed and successfully synthesized a series of $\text{Li}_3\text{YCl}_{6-x}\text{Br}_x$ compounds by ball milling and subsequent hot pressing. Their XRD patterns ($0 \leq x \leq 1.5$) are shown in Fig. 5a. With $x \leq 1.5$, a pure phase isostructural to LYC is obtained. The peak broadening observed at ~ 5.3 Å is greatly suppressed in the XRD patterns of $\text{Li}_3\text{YCl}_5\text{Br}$ and $\text{Li}_3\text{YCl}_{4.5}\text{Br}_{1.5}$; the Rietveld refinements of these compounds are shown in Supplementary Fig. 7. A noticeable impurity can be observed for $x = 2$, which can be indexed as a LYB phase (Supplementary Fig. 8). Because of the larger size of Br^- (1.96 pm) relative to that of Cl^- (1.81 pm)³², the lattice parameters of $\text{Li}_3\text{YCl}_{6-x}\text{Br}_x$ increase linearly with increasing Br doping (Supplementary Fig. 9a). Also, the peaks at ~ 3.9 and 5.0 Å become narrower and more symmetric with more Br doping, indicating a gradual increase in the coherent lengths of the in-plane ordering of the Y^{3+} vacancies. The $2c/a$ values of $\text{Li}_3\text{YCl}_{6-x}\text{Br}_x$ are compared in Supplementary Fig. 9b: with increasing Br content, the unit cell of $\text{Li}_3\text{YCl}_{6-x}\text{Br}_x$ lengthens along the c axis. The Young modulus and hardness of $\text{Li}_3\text{YCl}_{4.5}\text{Br}_{1.5}$ were measured via nanoindentation and compared with those of LYC. The results (Supplementary Table II) show that both the modulus and hardness of $\text{Li}_3\text{YCl}_{4.5}\text{Br}_{1.5}$ are lower than those of LYC, which is likely due to the higher polarizability of Br^- compared with Cl^- . Figure 5b shows the ionic conductivity of $\text{Li}_3\text{YCl}_{6-x}\text{Br}_x$ extracted from the EIS analysis of pellets hot pressed at 200 °C. The corresponding Nyquist plots are shown in Supplementary Figs. 10–13. The Arrhenius plots of the conductivity of samples with $x = 0.5, 1$ and 1.5 are almost identical at high temperature (>130 °C), showing an E_a of ~ 0.22 eV. Each of the samples shows a SIT at a different temperature, with T_c decreasing with increasing Br doping: 130 °C for LYC, 100 °C for $\text{Li}_3\text{YCl}_{5.5}\text{Br}_{0.5}$, 90 °C for $\text{Li}_3\text{YCl}_5\text{Br}$ and 70 °C for $\text{Li}_3\text{YCl}_{4.5}\text{Br}_{1.5}$. As a result, $\text{Li}_3\text{YCl}_{4.5}\text{Br}_{1.5}$ shows an ionic conductivity of

over 100 mS cm^{-1} at 80 °C and a σ_{RT} of 6.1 mS cm^{-1} (Fig. 5b). The same EIS measurements were also conducted on highly crystalline samples of LYC and $\text{Li}_3\text{YCl}_{4.5}\text{Br}_{1.5}$, which were synthesized at 500 °C over 5 h, formed into pellets and hot pressed at 200 °C. The same T_c values of 130 and 70 °C were again observed for LYC and $\text{Li}_3\text{YCl}_{4.5}\text{Br}_{1.5}$, respectively (Supplementary Figs. 14 and 15), despite the RT conductivity of the LYC sample synthesized at 500 °C being lower than that of the LYC sample hot pressed at 200 °C, which is likely a result of the difference in the crystallinity and Y distribution at the M2 and M3 sites^{24,25,28} due to the different synthesis protocols. We also collected the synchrotron diffraction data of the highly crystalline LYC at various temperatures; the Rietveld refinement data are presented in Supplementary Tables II–15. The ‘rocking’-to-‘breathing’ transition was also confirmed in highly crystalline LYC, demonstrated by the leaps in the intralayer Cl–Cl distances (S12 and S33) and bottleneck sizes (BN1 and BN3; Supplementary Fig. 16). If T_c could be further tuned down to RT, the resulting ionic conductivity would be above 20 mS cm^{-1} . However, on further increasing the Br doping, the LYB phase becomes the thermodynamic stable phase instead of LYC, and this change in long-range structure lowers the diffusivity and results in a lower conductivity in the samples tested.

To verify whether the SITs observed at lower temperatures for $\text{Li}_3\text{YCl}_{6-x}\text{Br}_x$ are indeed due to the ‘rocking’-to-‘breathing’ transition, synchrotron in situ XRD and ex situ variable-temperature ND analyses were conducted. The XRD data of $\text{Li}_3\text{YCl}_{4.5}\text{Br}_{1.5}$ collected in the same temperature range as the EIS study are shown in Supplementary Fig. 17a. Similarly to LYC, no obvious phase transition was identified during the entire heating and cooling cycle, other than the anisotropic thermal expansion and shrinkage of the unit cell (Supplementary Fig. 17b,c). The FDMs generated at 100 and 500 K are shown in Fig. 5c,d. Clear residual Li scattering length density can be seen on the two major lithium sites (6g and 6h) at both temperatures. No noticeable densities can be seen connecting these sites within the a – b plane (either in the $z = 0$ or $z = 0.5$ plane) at 100 K at a cut-off of $\sim 0.03 \text{ fm}^3 \text{ Å}^{-3}$ (Supplementary Fig. 18a). Further decreasing the residual density to the cut-off threshold of $\sim 0.02 \text{ fm}^3 \text{ Å}^{-3}$ shows the connection of residual

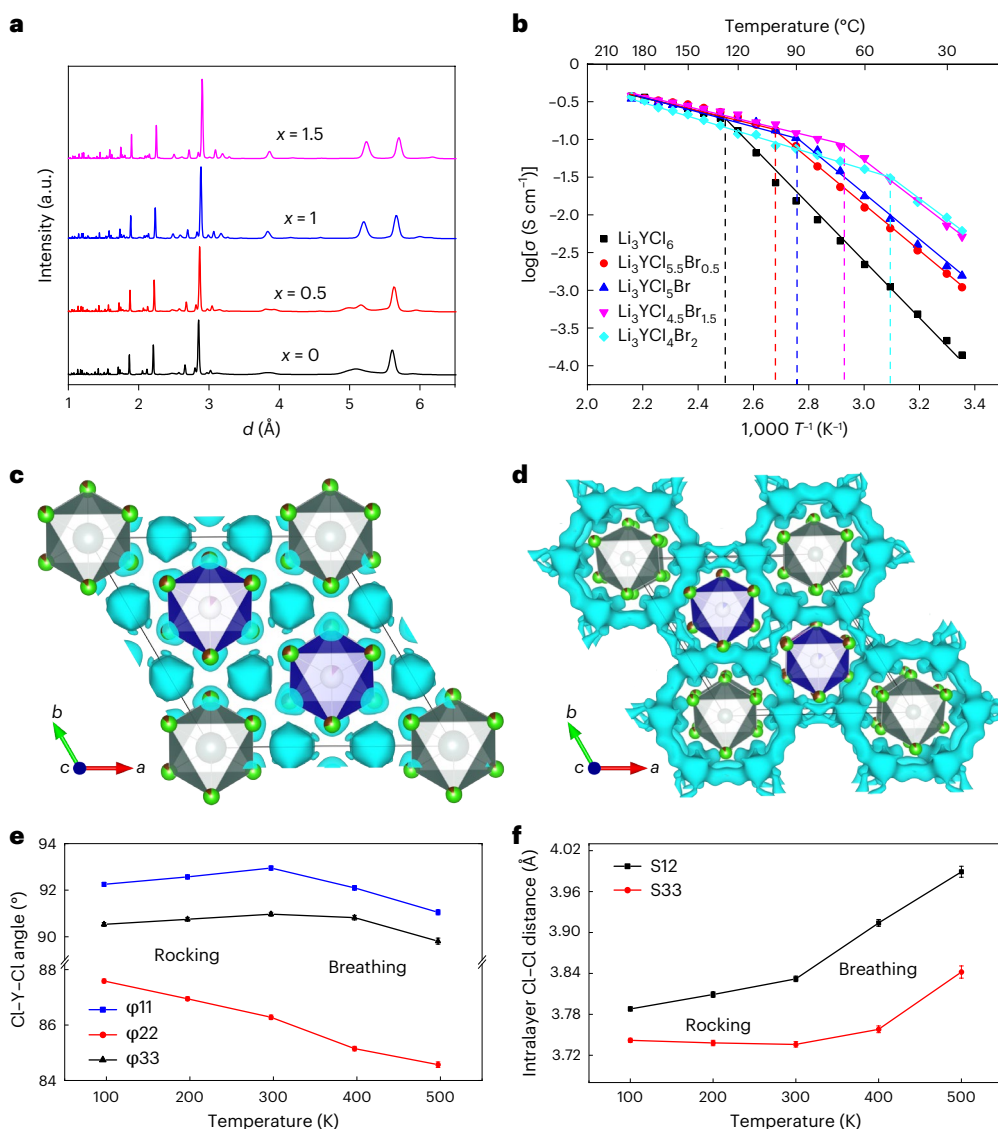


Fig. 5 | Activating 2D Li⁺ diffusion pathways within the *a*-*b* plane in Li₃YCl_{6-x}Br_x. **a**, Synchrotron XRD patterns ($\lambda = 0.24083$ Å) of LYC, Li₃YCl_{5.5}Br_{0.5}, Li₃YCl₅Br and Li₃YCl_{4.5}Br_{1.5}. Each sample crystallized with the *P3m1* structure with a minor LiCl impurity. **b**, Comparison of the Arrhenius plots of the conductivity of Li₃YCl_{6-x}Br_x. The vertical dashed lines are guides to the eyes. **c,d**, FDMs generated from ND data of YCl_{4.5}Br_{1.5} at 100 K with a threshold of -0.02 fm Å⁻³ (**c**) and at

500 K with a threshold of -0.015 fm Å⁻³ (**d**). **e**, Evolution of the Cl-Y-Cl angle in M1Cl₆ and M3Cl₆ octahedra with temperature. **f**, Evolution of the intralayer Cl-Cl distances S12 and S33 with temperature. The error bars represent the 'estimated standard deviations' of the corresponding refined parameters, which are a measure of the irreducible minimum in the uncertainty of the refined parameters during Rietveld refinement.

densities along the *c* axis and some residual densities can also be seen on the tetrahedral sites (Supplementary Fig. 18b). This implies that Li₃YCl_{4.5}Br_{1.5} is a one-dimensional (along the *c* axis) Li⁺ conductor at low temperature. In clear contrast, connected residual Li densities around the M1(Cl/Br)₆ octahedra (*z* = 0 plane) can be seen even at a high cut-off threshold (-0.02 fm Å⁻³) in the FDM generated at 500 K (Supplementary Fig. 19). These findings indicate that Li⁺ diffusion pathways are opened up within the *a*-*b* plane at 500 K. Similarly to LYC, the 2D diffusion channels are still not available in the *z* = 0.5 plane, as evidenced by the isolated residual Li densities even using a very low threshold (Supplementary Fig. 19b). To further understand why the 2D Li⁺ diffusion channels open up at a much lower temperature in Li₃YCl_{4.5}Br_{1.5} than in LYC, Rietveld refinements were carried out using both the variable-temperature ND and synchrotron XRD data. The refinements are shown in Fig. 5e,f and Supplementary Tables 6–10 for the ND data and in Supplementary Fig. 20 for the XRD data. A clear transition from the 'rocking' to 'breathing' type collective motion of

anions can be seen above ~340 K, as evidenced by the switch from the compression mode of M1(Cl/Br)₆ and M3(Cl/Br)₆ octahedra to the expansion (along the *c* axis) mode (ϕ_{11} and ϕ_{33} angles in Fig. 5e and Supplementary Fig. 20a,b). This transition drastically increases the S12 and S33 distances (intralayer (Cl/Br)-(Cl/Br), Fig. 5f and Supplementary Fig. 12c), resulting in the opening up of the diffusion bottleneck windows BN1 and BN3 (Supplementary Fig. 21). The electrochemical performance of all-solid-state cells employing a LiNi_{0.8}Mn_{0.1}Co_{0.1}O₂ (NMC811) cathode, Li₃YCl_{4.5}Br_{1.5} electrolyte and InLi alloy anode was next explored (Supplementary Fig. 22). The cells showed reasonably good capacity and rate performance, despite a relatively low stack pressure of ~20 MPa being used, benefiting from the high conductivity of Li₃YCl_{4.5}Br_{1.5}. Yet, more substantial engineering and optimization of the configuration and fabrication of the cells (for example, stack pressure, particle size of the solid electrolyte, and type and morphology of the cathode) are necessary to further improve the cycling and rate performances.

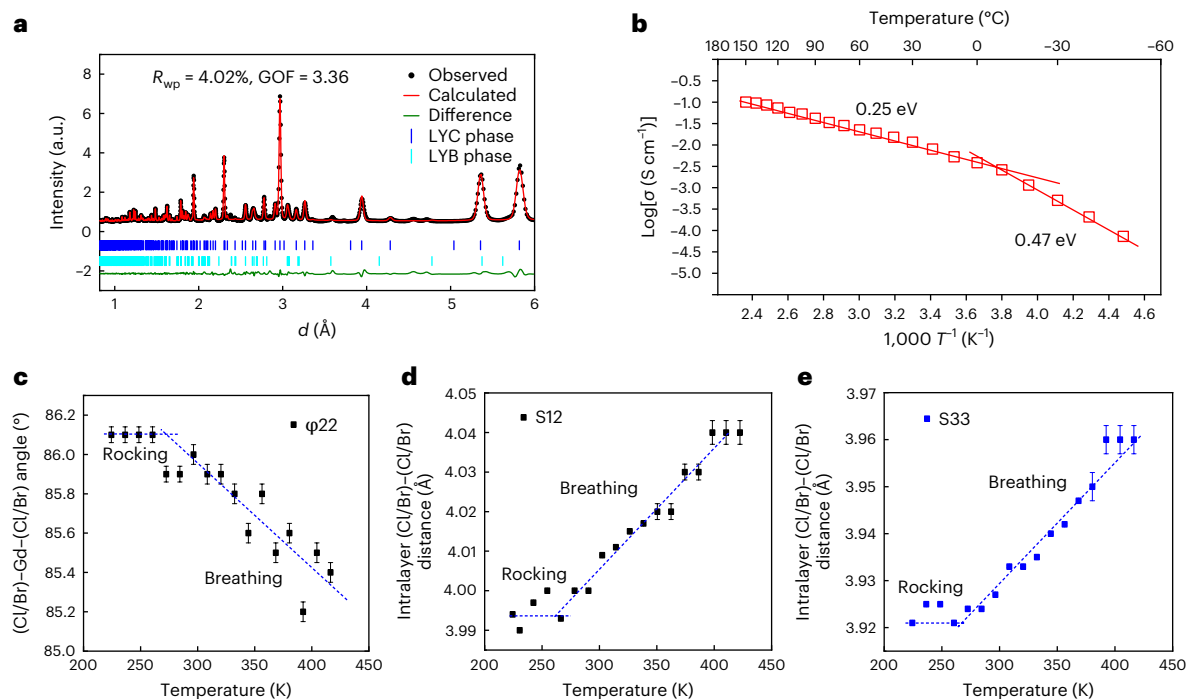


Fig. 6 | Design of $\text{Li}_3\text{GdCl}_3\text{Br}_3$ as a solid-state electrolyte. a, Rietveld refinement of $\text{Li}_3\text{GdCl}_3\text{Br}_3$ against the synchrotron XRD data. R_{wp} is the weighted residual. GOF is goodness of fit. The refinement shows that $\text{Li}_3\text{GdCl}_3\text{Br}_3$ mainly crystallizes in the LYC structure (85 wt%). **b**, Arrhenius plot of the conductivity of $\text{Li}_3\text{GdCl}_3\text{Br}_3$ for an extended low-temperature range. T_c is observed at -10°C . **c**, Evolution of the (Cl/Br)–Gd–(Cl/Br) angle ϕ_{22} in M_3Cl_6 octahedra with temperature.

d, e, Evolution of intralayer (Cl/Br)–(Cl/Br) distances S12 (**d**) and S33 (**e**) in the Gd layer with temperature. The error bars represent the ‘estimated standard deviations’ of the corresponding refined parameters, which are a measure of the irreducible minimum in the uncertainty of the refined parameters during Rietveld refinement.

Extending the Br doping limit in the $P\bar{3}m1$ structure

The above results confirm that substituting Br at a Cl site can better trigger the transition from the ‘rocking’ mode to the ‘breathing’ mode in LYC structures, resulting in a lower T_c of 70°C . However, the solid solubility limit of Br in $\text{Li}_3\text{YCl}_{6-x}\text{Br}_x$, is at $x = 1.5$. To further lower T_c to RT, it is necessary to push the Br doping limit higher. We therefore designed the $\text{Li}_3\text{GdCl}_{6-x}\text{Br}_x$ series of compounds with the expectation that the greater size of Gd^{3+} (0.938 pm)³² compared with Y^{3+} (0.9 pm) can promote the tolerance of Br mixing in the LYC structure. Thus, a series of $\text{Li}_3\text{GdCl}_{6-x}\text{Br}_x$ samples were synthesized and tested. Their XRD patterns and Arrhenius plots of the conductivity are shown in Supplementary Fig. 23a, b, respectively. Due to the larger size of Gd^{3+} compared with Y^{3+} , only with $x = 3$ is the LYC phase obtained with a minor LYB phase impurity. The Rietveld refinement is shown in Fig. 6a. The Nyquist plots of $\text{Li}_3\text{GdCl}_3\text{Br}_3$ collected at various temperatures are shown in Supplementary Fig. 24. Figure 6b shows the Arrhenius plot of the conductivity of $\text{Li}_3\text{GdCl}_3\text{Br}_3$, with a σ_{RT} of 11 mS cm^{-1} , which is the highest value so far to have been reported for halide-based lithium ionic conductors. Unfortunately, as Gd is an extremely absorbing element for ND, ND analysis was not conducted. However, $\text{Li}_3\text{GdCl}_3\text{Br}_3$ did show a lower T_c of -10°C , as evidenced by the very different E_a values above and below this temperature (Fig. 6b), which is similar to what was observed in the $\text{Li}_3\text{YCl}_{6-x}\text{Br}_x$ series. This result clearly demonstrates the success of the design strategy to lower T_c as an effective means to achieve very high conductivity at RT. In situ synchrotron XRD data were collected for $\text{Li}_3\text{GdCl}_3\text{Br}_3$ from 420 K down to 220 K. The diffraction patterns are shown in Supplementary Fig. 25 and the refinements in Fig. 6c–e. The angle ϕ_{22} clearly shows the transition from the ‘rocking’ to ‘breathing’ type collective motion of anions at -260 K (Fig. 6c), which is consistent with the results of the EIS analysis. As a result, the intralayer distances (S12 and S33) for the 2D diffusion channels within the a – b plane opens up (Fig. 6d, e).

Conclusion

This study has systematically revealed that the SIT in halide compounds with the LYC structure is triggered by the change in the collective motion of the anions, which not only explains the discrepancies between previous experimental and computational results but also offers an effective rational design strategy for electrolyte materials. $\text{Li}_3\text{YCl}_x\text{Br}_{6-x}$ and $\text{Li}_3\text{GdCl}_3\text{Br}_3$ were designed following the strategy to reduce T_c . Lower T_c values of 70°C and -10°C were realized and high RT conductivities of 6.1 and 11 mS cm^{-1} were achieved in $\text{Li}_3\text{YCl}_{4.5}\text{Br}_{1.5}$ and $\text{Li}_3\text{GdCl}_3\text{Br}_3$, respectively. This is a successful example of a rational material design guided by the structure–property relationship revealed by in-depth crystal structure characterization using synchrotron and neutron scattering techniques. The discovery of highly conductive halide solid electrolytes offers new options for the development of ALSOLIBs and the design of ionic conductors.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41557-024-01634-6>.

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Methods

Material synthesis

$\text{Li}_3\text{YCl}_{6-x}\text{Br}_x$ ($x = 0, 0.5, 1, 1.5$) were synthesized via high-energy ball milling and subsequent heat treatment. Anhydrous LiCl (>99%, Sigma-Aldrich), LiBr (>99%, Sigma-Aldrich) and anhydrous YCl_3 (99.99%, Sigma-Aldrich) as starting materials were weighed in the desired molar ratio and ball milled with zirconia jars (50 ml) and balls in a planetary ball mill (PM200, Retsch) at 500 r.p.m. for 5 h. To prepare hot-pressed LYC, $\text{Li}_3\text{YCl}_{5.5}\text{Br}_{0.5}$, $\text{Li}_3\text{YCl}_5\text{Br}$ and $\text{Li}_3\text{YCl}_{4.5}\text{Br}_{1.5}$, the ball-milled powders were pressed into pellets, heated under 1.5 metric tonne to 200 °C and held at this temperature for 10 min. All of the processes were carried out under an Ar atmosphere. $\text{Li}_3\text{GdCl}_{6-x}\text{Br}_x$ were synthesized following a similar protocol using anhydrous LiCl (>99%, Sigma-Aldrich), LiBr (>99%, Sigma-Aldrich), anhydrous GdCl_3 (≥99%, Strem) and anhydrous GdBr_3 (≥99%, Strem) in the desired molar ratio.

Material characterization

Laboratory XRD data were collected with a D8 Advance X-ray diffractometer (Bruker AXS) using Mo radiation ($\lambda_{\text{MoK}\alpha 1} = 0.7093 \text{ \AA}$). High-resolution synchrotron XRD spectra were collected at beam line 17-BM of the Advanced Photon Source at Argonne National Laboratory and at the 28ID-2 XPD beamline of the National Synchrotron Light Source II at Brookhaven National Laboratory.

ND and nPDF analysis

Neutron total scattering data were collected at the Spallation Neutron Source NOMAD beamline³³. Powder samples (~0.3 g) were loaded into 3 mm thin-walled quartz capillaries for measurement. The collected scattering data were background subtracted from the data acquired from empty 3 mm quartz capillaries and then normalized by the scattering from a 6 mm vanadium rod to correct for the incident beam profile and detector efficiency. Bragg diffraction data were reduced in a bank-by-bank manner (data from detector banks centred at 31°, 67°, 122° and 154° were used for Rietveld refinement). For the nPDF data reduction, the one-frame structure function $S(Q)$ was Fourier transformed into the reduced pair distribution function using the following equation, where $G(r)$ is the reduced pair distribution function, $g(r)$ is the atomic pair distribution function, r is the atomic pair distance, ρ is the atomic number density, Q is the momentum transfer and $Q_{\text{max}} = 35 \text{ \AA}^{-1}$:

$$G(r) = [1 - g(r)]r = \frac{1}{2\pi^2\rho_0} \int_{Q_{\text{min}}}^{Q_{\text{max}}} Q[S(Q) - 1] \sin(Qr) dQ$$

Structure refinement

Neutron. Structure refinements using neutron Bragg diffraction data were carried out using TOPAS (v6) software³⁴. Time-of-flight (TOF) data were converted to d spacing using the conventional second-order polynomial $\text{TOF} = \text{ZERO} + \text{DIFC} \times d + \text{DIFA} \times d^2$, where ZERO is a constant, DIFC is the diffractometer constant and DIFA is an empirical term to correct for sample displacements and absorption-induced peak shifts. DIFC was determined from the refinement of a standard National Institute of Standards and Technology (NIST) Si-640e dataset and held fixed, while ZERO and DIFA were refined to account for sample displacement and wavelength-dependent effects. The moderator-induced line profile was modelled using a modified Ikeda–Carpenter–David function^{35,36}. The Lorentz factor was corrected by multiplying by d^4 (ref. 37). Absorption correction was carried out using the empirical Lobanov formula (assuming each detector bank has an average scattering angle)³⁸.

Rietveld refinement using X-ray Bragg diffraction. The fundamental peak profile approach was used to describe the instrument peak profile. The peaks broadening from size and microstrain were also included during the refinement. Sequential structure refinements were carried out using TOPAS (v6).

Electrochemical measurements

Ionic conductivity was measured by EIS using an electrochemical impedance analyser (VMP3, BioLogic) and a custom-built electrochemical cell. Typically, ~1 g $\text{Li}_3\text{YCl}_{6-x}\text{Br}_x$ powder was pressed into a pellet with a diameter of 1.27 cm at a pressure of 100 bar. Two stainless-steel rods were used as the ion-blocking electrodes. EIS data were collected in the temperature range of 25 °C to 200 °C in a frequency range of 1 MHz to 1 Hz and with AC amplitude of 50 mV.

A battery configuration similar to that previously reported was used for the all-solid-state battery assembly¹². Thus, NMC811, LYC and Super P were mixed in a mass ratio of 65:30:5 as the cathode composite. Then, 120 mg $\text{Li}_3\text{YCl}_{4.5}\text{Br}_{1.5}$ was pressed into a pellet ($d = 0.5 \text{ inch}$) in a poly(methyl methacrylate) sleeve at 294 MPa as the separator. Around 10 mg cathode composite was spread over one side of the pellet and pressed at 294 MPa. InLi alloy was then attached to the other side of the pellet to serve as the anode. Stainless-steel rods were used as current collectors. Cycling tests were performed on the all-solid-state cells in galvanostatic mode between 2.52 and 4.3 V versus Li^+/Li at RT.

Computation methods

First-principles computation. All density functional theory (DFT) calculations were carried out using the Vienna Ab initio Simulation Package³⁹ within the projector augmented wave approach⁴⁰. The generalized gradient approximation was used with the Perdew–Burke–Ernzerhof functional⁴¹ for most of the energy calculations. The plane-wave energy cut-off and k -point density used in the calculations were consistent with that of the Materials Project⁴².

Structural ordering. The LYC and $\text{Li}_3\text{YCl}_{6-x}\text{Br}_x$ samples are well indexed to the hexagonal close-packed structure with space group $P3m1$, where Li partially occupies the octahedral $6h$ and $6g$ sites. In $\text{Li}_3\text{YCl}_{6-x}\text{Br}_x$, Br atoms partially occupy the Cl $6i$ sites. Based on the experimentally refined structure, we enumerated all configurations in a unit cell of LYC and generated 50 symmetrically distinctive structures in a unit cell of $\text{Li}_3\text{Y}(\text{Cl}/\text{Br})_6$ by randomizing the disordered occupancies. Then we included structural relaxation in the DFT calculations, and the lowest-energy structure was identified as the ground-state structure for further calculations. It should be noted that the same uniform Y sublattice was used in all our structural ordering to reduce the Coulombic repulsion.

AIMD simulations. We performed AIMD simulations using $1 \times 1 \times 2$ supercell models with lattice parameters larger than 10 Å in each lattice direction using the NVT ensemble with the Nosé–Hoover thermostat⁴³. Non-spin mode, a time step of 2 fs and a Γ -centred $1 \times 1 \times 1$ k -point grid were used. In each simulation, structures were heated from 100 K to the target temperature (350–900 K) at a constant rate during a period of 2 ps. The lithium ionic conductivity (σ) was calculated according to the Nernst–Einstein relation:

$$\sigma = \frac{N}{V} \frac{q^2}{k_B T} D = \frac{q^2}{Vk_B T} \frac{\text{TMSD}(\Delta t)}{2\Delta t}$$

where N is the number of mobile carriers, D is the diffusivity, Δt is the time interval, d is the diffusion dimensionality, which is 3 in our simulation, V is the volume of the model, q is the charge of the carrier, k_B is the Boltzmann constant and T is the temperature. TMSD is the total mean square displacement and represents the total diffusion of all lithium ions in the material. The Arrhenius relation was used to calculate E_a and to extrapolate the ionic conductivity to the desired temperature:

$$\sigma T = \sigma_0 \exp\left(\frac{-E_a}{k_B T}\right)$$

where σ_0 is the pre-exponential factor. Given that ion hopping is a stochastic process, the statistical deviations of conductivity were

evaluated according to the number of hopping events in our previous report⁴⁴. The total duration of the AIMD simulations was in the range of 100–400 ps until the ionic diffusivity converged with a relative standard deviation in the range of 20–40%.

Data availability

All of the data that support the findings of this work are available within the paper and its Supplementary Information. Source data are provided with this paper.

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Author contributions

Z.L. and H.C. conceived the idea and designed the experiments. Z.L. conducted the synthesis, electrochemical tests and some of the characterizations. P.-H.C. and J.L. performed the ND characterizations and analyses. S.W. performed the computer simulations under the supervision of Y.M. S.S. and S.C. contributed to the processing of the materials. M.L., Z.L. and S.X. performed mechanical property measurements and analyses. Z.L., H.C., J.L. and S.W. wrote the paper. All authors reviewed and revised the paper.

Competing interests

The authors declare no competing interests.

Additional information

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